AI / Data Science

Task Report — Day 06

Advanced Pipelines · NLP · Anomaly Detection · Recommendation Engine

DATE
17 June 2026

DOMAIN
Artificial Intelligence / Data Science

DATASETS
Wine Quality (1599 rows) + NLP Text Docs (15 samples)

TASKS
10 / 10 Tasks Completed

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1 Task Overview

Day 06 — All 10 tasks completed

Day 06 focuses on advanced AI and data science concepts that build directly on the skills from Days 01–05. The 10 tasks cover the full spectrum of modern machine learning workflows — from raw text processing all the way to anomaly detection, automated pipelines, and intelligent recommendation systems.

#	Task	Implementation	Status
1	End-to-end ML pipeline	StandardScaler + SelectKBest + 3 models	**Done**
2	Document classification system	TF-IDF + Multinomial Naive Bayes	**Done**
3	Text & info extraction	Keyword extraction via TF-IDF scores	**Done**
4	Advanced text preprocessing	Regex + NLTK stopwords + stemming	**Done**
5	Anomaly detection model	IsolationForest (contamination=0.05)	**Done**
6	Recommendation engine	Cosine similarity on wine features	**Done**
7	Predictive analytics	Wine quality prediction pipeline	**Done**
8	Compare ML approaches	LR vs RF vs Gradient Boosting with CV	**Done**
9	Automated insights from dataset	Auto-printed stats and business metrics	**Done**
10	Technical documentation	README + PDF report	**Done**

2 Datasets Used

Dataset 1 — Red Wine Quality (Numeric)

Source: UCI Wine Quality Dataset **File:** wine.csv (copied from Day05) **Separator:** Semicolon (;)
Rows: 1,599 **Columns:** 12

Column	Description	Used For
fixed acidity	Tartaric acid content	Pipeline, Anomaly
volatile acidity	Acetic acid (too high = vinegar)	Pipeline, Anomaly, Rec
citric acid	Adds freshness to wine	Pipeline
residual sugar	Sugar after fermentation	Pipeline
chlorides	Salt content	Pipeline
free sulfur dioxide	Prevents microbial growth	Pipeline
total sulfur dioxide	Total SO2 present	Pipeline
density	Density of the wine	Pipeline
pH	Acidity level (0-14)	Pipeline
sulphates	Wine preservative	Pipeline, Anomaly
alcohol	Alcohol percentage	Pipeline, Anomaly, Rec
quality	Score 3-8 → binary target (>=6=Good)	Target

Dataset 2 — NLP Text Documents (Inline)

Source: Created inline in analysis.py — no download required **Total Docs:** 15 **Categories:** Tech (5), Sports (5), Food (5)

The 15 text documents were deliberately crafted to represent 3 distinct domains. They serve as the training and testing data for the document classification system and keyword extraction tasks. Having 5 documents per category ensures balanced representation and allows TF-IDF to learn domain-specific vocabulary.

Category	Example Document
Tech	Python is a powerful programming language for data science and machine learning
Sports	The football team won the championship after an incredible final performance
Food	Italian pasta dishes are loved worldwide for their rich and authentic flavors

3 Advanced Text Preprocessing

Task 4 — Cleaning raw text for NLP

Before any NLP model can process text, the raw input must be cleaned and normalised. Raw text contains capital letters, punctuation, common filler words (called stopwords), and different forms of the same word (e.g. 'running', 'runs', 'ran'). Preprocessing removes all of this noise so the model can focus on the meaningful content words.

The 4-Step Preprocessing Pipeline

Step 1 **Lowercase**

Converts all characters to lowercase. This ensures 'Python' and 'python' are treated as the same word. Without this, the model would see them as two different tokens.

```
# Lowercase
text.lower()
```

Step 2 **Punctuation & Number Removal**

Uses a regular expression to remove everything that isn't a letter or space. This strips commas, periods, numbers, hyphens and other special characters that add no semantic meaning to the text.

```
# Punctuation & Number Removal
re.sub(r'[^a-z\s]', '', text)
```

Step 3 **Stopword Removal**

Stopwords are common English words like 'is', 'the', 'a', 'for', 'and' — words that appear in every document and carry no useful information for classification. NLTK provides a list of 179 English stopwords which are filtered out.

```
# Stopword Removal
[w for w in tokens if w not in stop_words]
```

Step 4 **Stemming (Porter Stemmer)**

Stemming reduces words to their base root form. For example: 'learning' → 'learn', 'science' → 'scienc', 'powerful' → 'power'. This means the model treats different grammatical forms of the same word as identical tokens.

```
# Stemming (Porter Stemmer)
stemmer.stem(word)
```

Before & After Example

Stage	Text
Original	Python is a powerful programming language for data science and machine learning
After lowercase	python is a powerful programming language for data science and machine learning

After punctuation rm	python is a powerful programming language for data science and machine learning
After stopword rm	python powerful programming language data science machine learning
After stemming	python power program languag data scienc machin learn

Insight After 4 preprocessing steps, 8 content-rich tokens remain from the original 16-word sentence. The noise is removed and the model now has clean, comparable, root-form tokens to work with.

4 Document Classification System

Task 2 — TF-IDF + Naive Bayes

What is TF-IDF?

TF-IDF stands for Term Frequency–Inverse Document Frequency. It is a numerical method that converts text documents into vectors of numbers that a machine learning model can process. It works by calculating two scores for every word in every document:

- **TF (Term Frequency):** How often a word appears in a specific document. Words that appear more often get a higher score.
- **IDF (Inverse Document Frequency):** How rare a word is across all documents. Common words like 'the' get a low IDF score; rare domain-specific words like 'blockchain' get a high IDF score.
- **TF-IDF = TF × IDF:** A word scores highest when it appears frequently in one document but rarely in others — making it a good identifier for that document's topic.

What is Multinomial Naive Bayes?

Naive Bayes is a probabilistic classifier based on Bayes' theorem. It works by calculating the probability that a document belongs to each category, given its word frequencies. The 'Multinomial' variant is specifically designed for text classification where features are word counts or TF-IDF scores. Despite its 'naive' assumption that words are independent of each other, it performs surprisingly well on text classification tasks.

Pipeline Architecture

```
nlp_pipeline = Pipeline([
    ('tfidf',  TfidfVectorizer()),   # Step 1: convert text to numbers
    ('model',  MultinomialNB())      # Step 2: classify using Naive Bayes
])

# Split: 10 training docs, 5 test docs
nlp_pipeline.fit(X_train_nlp, y_train_nlp)
nlp_pred = nlp_pipeline.predict(X_test_nlp)
```

Classification Results

Class	Precision	Recall	F1-Score	Test Samples
Tech	0.33	1.00	0.50	1
Sports	0.00	0.00	0.00	2
Food	1.00	1.00	1.00	2
Overall Accuracy	**—**	**—**	**0.60**	**5**

Note The model achieved 60% accuracy (3 out of 5 test documents correct). Food documents were classified perfectly (precision and recall both 1.0). Sports was the hardest category. IMPORTANT: With only 5 test samples, a single misclassification drops accuracy by 20%. This result is expected and normal for a dataset of this size — with more documents, accuracy would increase significantly.

5 Keyword Extraction

Task 3 — Top domain vocabulary per category

After TF-IDF vectorisation, the average TF-IDF score for each word across all documents in a category was computed. The words with the highest average scores are the most representative keywords for that category — they appear frequently in that domain but rarely in others.

Rank	Tech Keywords	Sports Keywords	Food Keywords
1	technolog	player	flavor
2	learn	team	dish
3	threat	perform	fresh
4	connect	footbal	celebr
5	grow	final	chocol

Note: keywords are shown in stemmed form as processed by Porter Stemmer.

Insight The keywords are clearly domain-specific and non-overlapping — 'technolog' only appears in Tech docs, 'player' and 'footbal' only in Sports, and 'flavor' and 'chocol' only in Food. This demonstrates that TF-IDF successfully identifies the distinguishing vocabulary of each category, which is exactly what keyword extraction aims to achieve.

6 End-to-End ML Pipeline

Task 1, 7, 8 — Advanced sklearn Pipeline with feature selection

An end-to-end machine learning pipeline automatically chains together multiple processing steps so that data flows through each stage in order. This is the professional way to build ML systems — it prevents data leakage and makes the workflow reproducible and easy to deploy.

Pipeline Architecture (3 stages)

Stage	Component	What it does
1	StandardScaler	Scales all features to mean=0, std=1 so no feature dominates due to different units
2	SelectKBest (k=8)	Selects the top 8 most statistically significant features using f_classif (ANOVA F-test)
3	ML Model	Trains the classifier on the scaled, selected features

```
def advanced_pipeline(model, k=8):
    return Pipeline([
        ('scaler',   StandardScaler()),           # Stage 1: scale
        ('selector', SelectKBest(f_classif, k=8)),# Stage 2: select top 8 features
        ('model',    model)                       # Stage 3: train model
    ])
```

7 Pipeline Model Comparison

Task 8 — LR vs RF vs Gradient Boosting

Three different classification models were tested inside the advanced pipeline. Each model was evaluated using both a single test set and 5-fold cross validation to get a more reliable estimate of real-world performance.

Models Explained

Logistic Regression	A statistical model that finds a linear decision boundary between classes. It's fast, interpretable, and works well when features are well-scaled (which our StandardScaler handles). Best suited for linearly separable problems.
Random Forest	An ensemble of decision trees where each tree votes on the output. More powerful than a single decision tree as it reduces overfitting through averaging. Works well with both linear and non-linear relationships.
Gradient Boosting	Builds trees sequentially where each new tree corrects the errors of the previous ones. Generally the most accurate of the three but also the slowest to train. Particularly good at capturing complex patterns in data.

Results

Model	Test Accuracy	CV Mean Accuracy	Verdict
Logistic Regression	0.73	0.74	Best CV score (tied)
Random Forest	0.79	0.74	Best test accuracy
Gradient Boosting	0.76	0.73	Slightly lower CV

Insight Logistic Regression and Random Forest tied at 0.74 CV accuracy. Random Forest achieved the best single test run (0.79). All three models perform similarly, suggesting that the wine quality features have a natural accuracy ceiling around 74-79% for this binary classification task regardless of which algorithm is used.

8 Anomaly Detection

Task 5 — IsolationForest on wine dataset

What is Anomaly Detection?

Anomaly detection (also called outlier detection) is an unsupervised machine learning technique that identifies data points which behave significantly differently from the rest of the dataset. Unlike classification, anomaly detection does not require labelled data — it learns what 'normal' looks like and flags anything that deviates too far from that pattern.

What is IsolationForest?

IsolationForest is an algorithm that isolates anomalies by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of that feature. The logic is simple: anomalies are few and different, so they require fewer splits to be isolated compared to normal data points. It builds many random trees and measures how many splits it takes to isolate each data point — fewer splits = more likely to be an anomaly.

Configuration & Results

```
iso = IsolationForest(
    contamination=0.05,  # expect 5% of data to be anomalies
    random_state=42
)
df['anomaly'] = iso.fit_predict(X_anomaly)
# Output: 1 = Normal, -1 = Anomaly
```

Setting / Result	Value	Explanation
Features Used	4	alcohol, volatile acidity, sulphates, quality
contamination	0.05	Tells the model to expect ~5% of data as anomalies
Normal Wines	1,520	95.1% of wines behave as expected
Anomaly Wines	79	4.9% of wines have unusual feature combinations
Output format	1 / -1	1 = Normal, -1 = Anomaly

Insight 79 wines (4.9%) were flagged as anomalies — wines with unusual combinations of alcohol level, volatile acidity, and sulphates that don't match the typical pattern. In a real winery scenario, these could represent wines with production defects, mislabelled batches, or simply rare high/low quality outliers worth investigating further.

9 Recommendation Engine

Task 6 — Content-based filtering with cosine similarity

What is a Recommendation Engine?

A recommendation engine is a system that suggests relevant items to a user based on some criteria. There are two main types: collaborative filtering (based on what similar users liked) and content-based filtering (based on the features of the items themselves). We implemented a content-based approach — finding wines with the most similar feature profiles to a reference wine.

What is Cosine Similarity?

Cosine similarity measures how similar two vectors are by calculating the cosine of the angle between them. A score of 1.0 means the vectors point in exactly the same direction (identical feature profiles), while 0 means they are completely unrelated. It is widely used in recommendation systems because it works well even when the magnitude of values differs.

```
from sklearn.metrics.pairwise import cosine_similarity

# Scale wine features so no single feature dominates
wine_scaled = StandardScaler().fit_transform(wine_features)

def recommend_wines(wine_idx, n=3):
    sims    = cosine_similarity([wine_scaled[wine_idx]], wine_scaled)[0]
    top_idx = sims.argsort()[::-1][1:n+1]  # skip index 0 (the wine itself)
    return top_idx, sims[top_idx]
```

Results — Recommendations for Wine #0

Role	Wine #	Similarity	Quality	Alcohol
Reference Wine	**#0**	**—**	**5**	**9.4%**
Recommendation 1	#4	1.000	5	9.4%
Recommendation 2	#5	0.992	5	9.4%
Recommendation 3	#28	0.979	5	9.4%

Insight Wine #4 received a perfect similarity score of 1.0 — meaning it has an identical feature profile to Wine #0 (same alcohol, acidity, sulphates, etc.). This makes sense as wines in the dataset often share similar profiles. All 3 recommendations share the same quality score (5) and alcohol level (9.4%), confirming the engine is finding genuinely similar wines rather than random matches.

10 Visualizations & Automated Insights

Task 9, 10

Chart 1	**Category Distribution** (Bar Chart) -> chart1_category_distribution.png
Insight	Equal distribution of 5 documents per category ensures balanced training data for the classifier.
Chart 2	**Top Keywords per Category** (Horizontal Bar (3 subplots)) -> chart2_top_keywords.png
Insight	TF-IDF scores clearly separate domain vocabulary — Tech, Sports, and Food keywords are non-overlapping, confirming the preprocessing worked correctly.
Chart 3	**Anomaly Detection** (Scatter Plot) -> chart3_anomaly_detection.png
Insight	79 anomaly wines (red dots) are visible as outliers in the alcohol vs volatile acidity feature space, forming clusters at the extremes of both axes.
Chart 4	**Pipeline Model Comparison** (Bar Chart) -> chart4_pipeline_comparison.png
Insight	All 3 pipeline models achieved similar CV accuracy (0.73-0.74), confirming a natural performance ceiling for this dataset.
Chart 5	**Recommendation Engine Output** (Bar Chart) -> chart5_recommendations.png
Insight	Wine #4 received a perfect similarity score of 1.0 — the recommendation engine correctly identified the most similar wines in the dataset.

Automated Business Insights (Task 9)

Metric	Value	Meaning
Total wines analysed	1,599	Full dataset processed
Good wine rate	53.5%	Slightly more good wines than bad
Anomaly rate	4.9%	79 wines with unusual feature profiles
Avg alcohol — good wines	10.86%	Higher alcohol in better-quality wines
Avg alcohol — bad wines	9.93%	Lower alcohol in poor-quality wines
Best CV model	LR / RF (0.74)	Both tied on cross-validation score

NLP classifier accuracy	0.60	3/5 test docs correct (small test set)

11 Key Findings & Conclusion

Key Findings

- TF-IDF + Naive Bayes successfully classified documents — Food category achieved perfect precision and recall
- Text preprocessing (lowercase, stopword removal, stemming) reduced noise and improved model token quality
- Advanced sklearn Pipeline with SelectKBest reduces features from 11 to 8, improving generalisation
- All 3 pipeline models achieved similar CV accuracy (~0.74) — the wine dataset has a natural accuracy ceiling
- IsolationForest correctly identified 79 anomaly wines (4.9%) with unusual feature combinations
- Cosine similarity recommendation engine found Wine #4 with a perfect similarity score of 1.0
- Good wines have consistently higher alcohol content (10.86% vs 9.93%) than bad wines
- NLP accuracy of 60% is expected and normal with only 5 test samples — more data would improve this significantly

What Was Learned

Skill / Concept	What was done
Advanced ML Pipelines	Learned to chain StandardScaler + SelectKBest + model in one reusable pipeline
TF-IDF Vectorisation	Converted raw text documents into numerical feature vectors for classification
Naive Bayes Classification	Learned probabilistic text classification using word frequency distributions
Text Preprocessing	Applied 4-step NLP preprocessing: lowercase, regex cleaning, stopwords, stemming
IsolationForest	Used unsupervised anomaly detection to find outlier wines without labelled data
Cosine Similarity	Built a content-based recommendation engine using vector similarity
NLTK Library	Used NLTK for stopword corpus and Porter Stemmer for token normalisation
Automated Insights	Programmatically generated business-relevant statistics from raw dataset

Task Status: Complete — 10 / 10 Tasks Done
All Day 06 objectives were successfully implemented: advanced ML pipelines, document classification with TF-IDF and Naive Bayes, keyword extraction, 4-step text preprocessing, IsolationForest anomaly detection, cosine similarity recommendation engine, automated business insights, and 5-chart dashboard. Project fully documented and pushed to GitHub.